

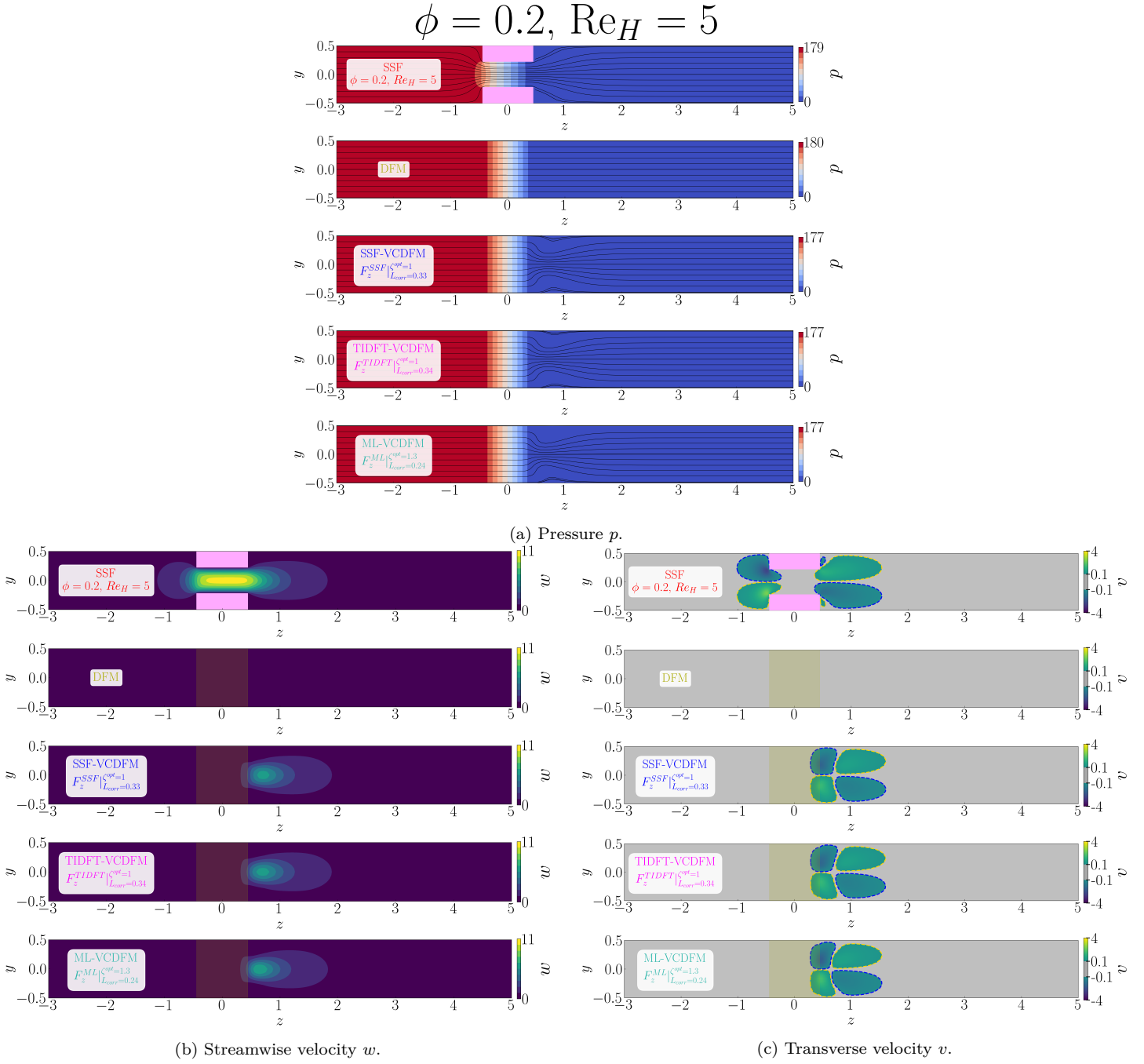


Figure D.130: Flow organization in the streamwise  $(z, y)$ -plane for the  $1 \times 1$  SRG at  $\phi = 0.2$ ,  $\text{Re}_H = 5$ , and  $\alpha = 0^\circ$ . The blocks show (a) pressure  $p$  with streamlines, (b) streamwise velocity  $w$ , and (c) transverse velocity  $v$ . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

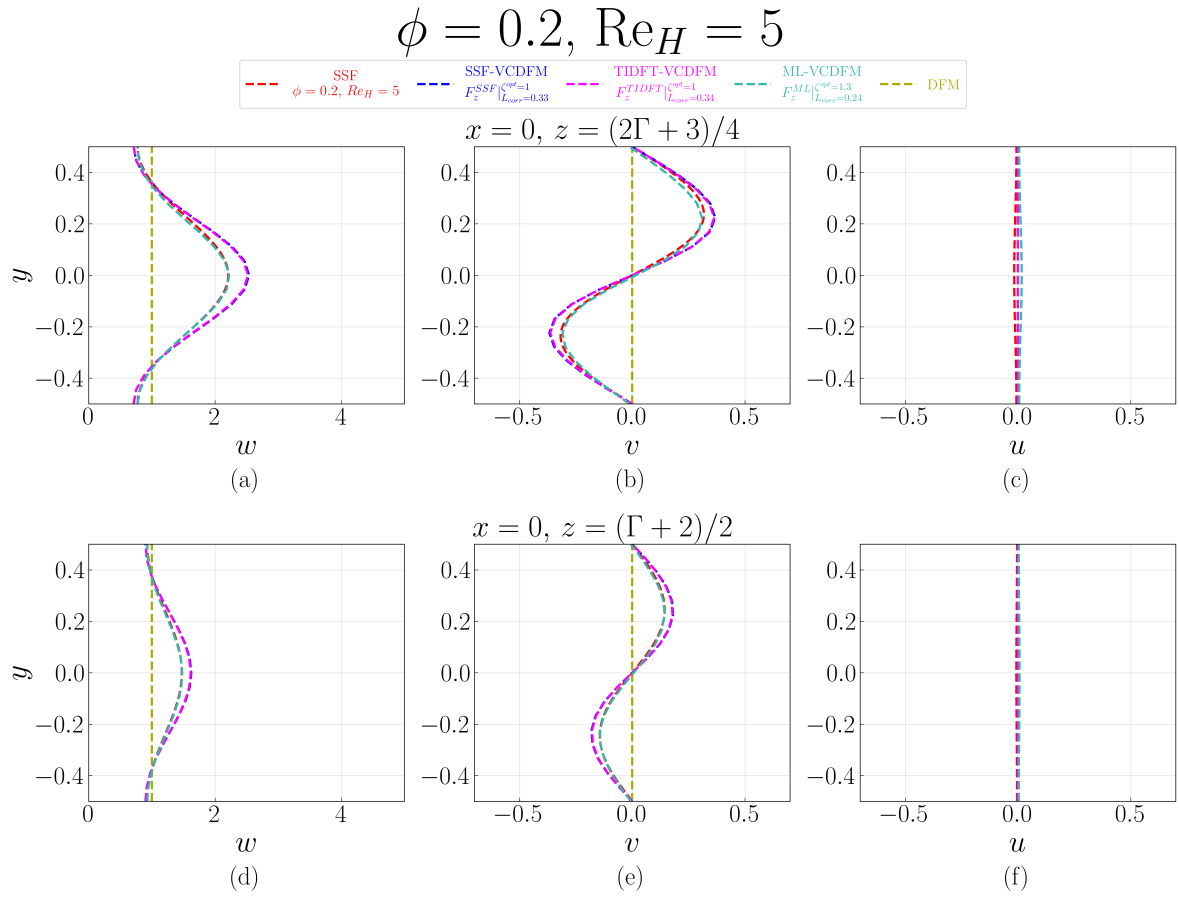


Figure D.131: Velocity profiles at  $x = 0$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 5$ , comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of  $w(y)$ ,  $v(y)$ , and  $u(y)$  are shown at  $z = (2\Gamma + 3)/4$  in (a–c) and  $z = (\Gamma + 2)/2$  in (d–f).

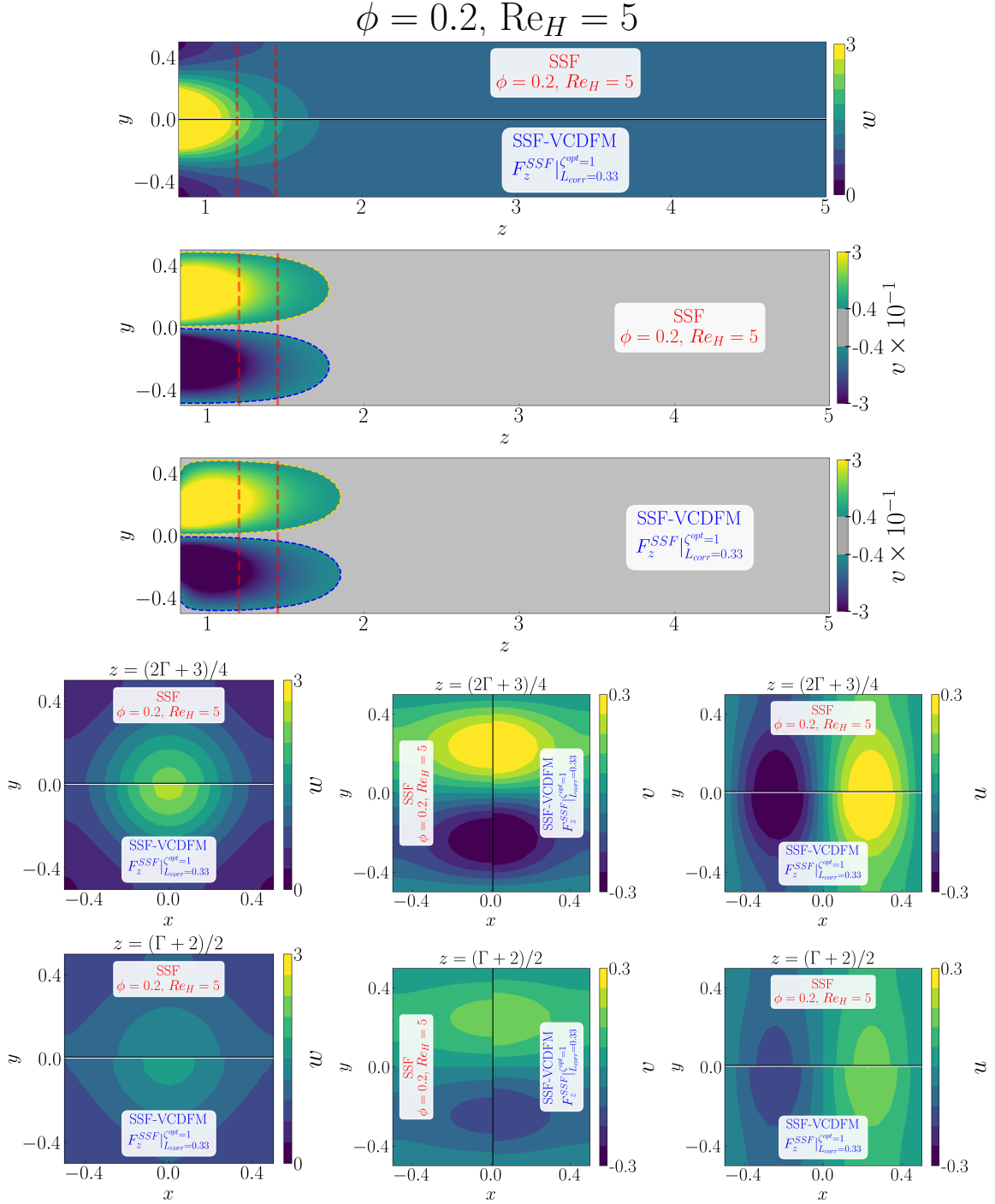


Figure D.132: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using  $F_z^{\text{SSF}}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 5$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (2\Gamma + 3)/4$  in the first row and  $z = (\Gamma + 2)/2$  in the second row.

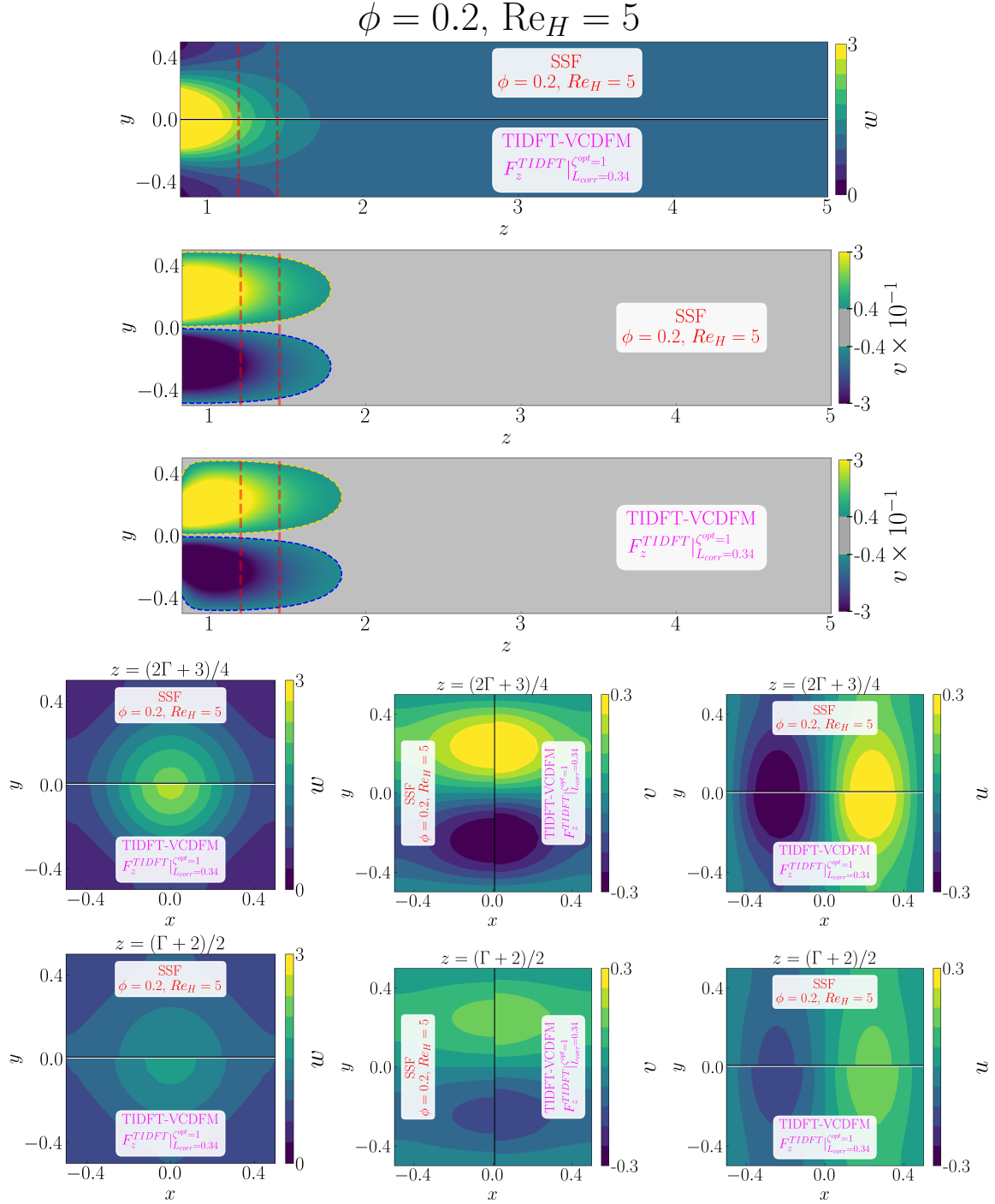


Figure D.133: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using  $F_z^{TIDFT}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $Re_H = 5$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (2\Gamma + 3)/4$  in the first row and  $z = (\Gamma + 2)/2$  in the second row.

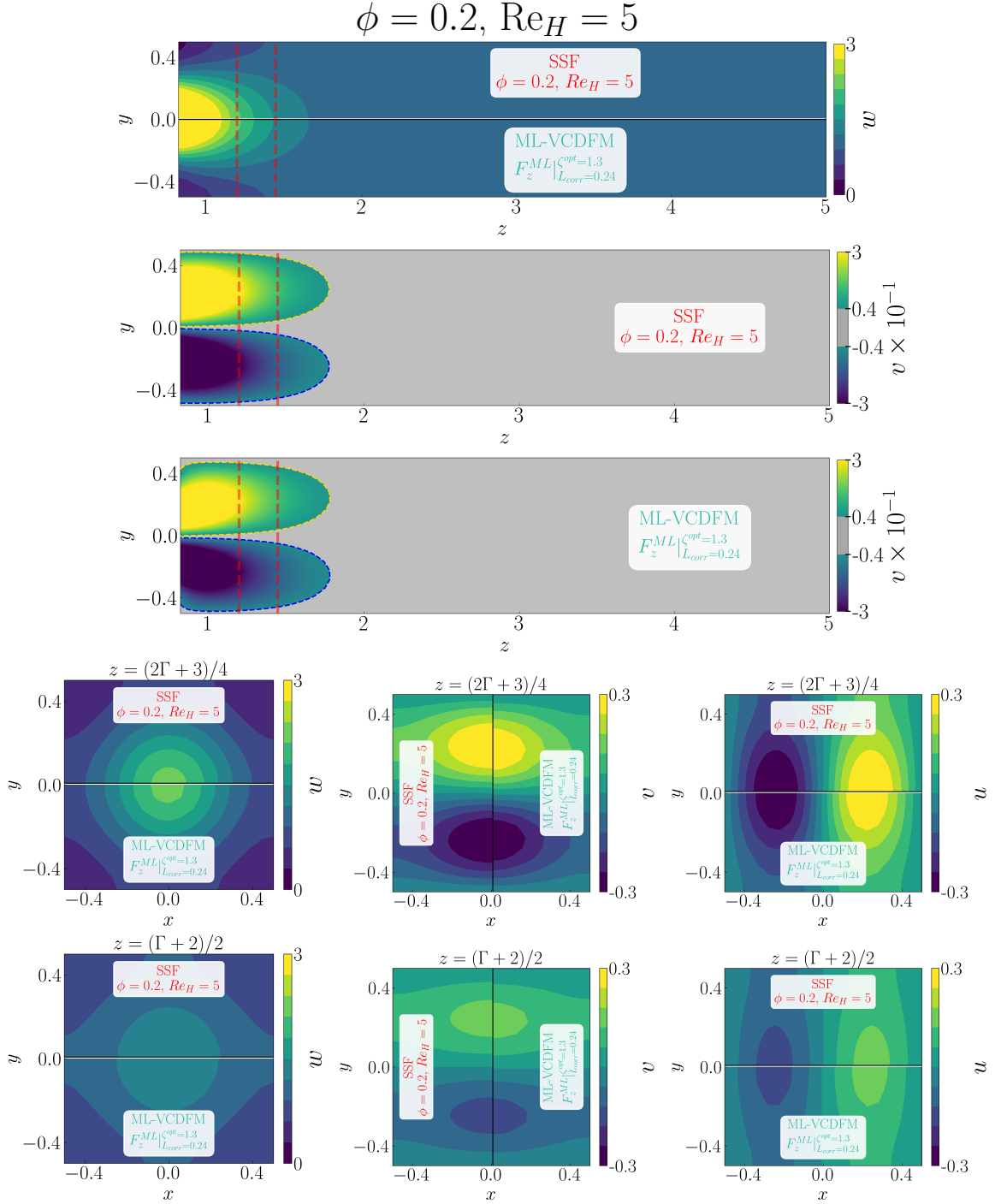


Figure D.134: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using  $F_z^{\text{ML}}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 5$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (2\Gamma + 3)/4$  in the first row and  $z = (\Gamma + 2)/2$  in the second row.

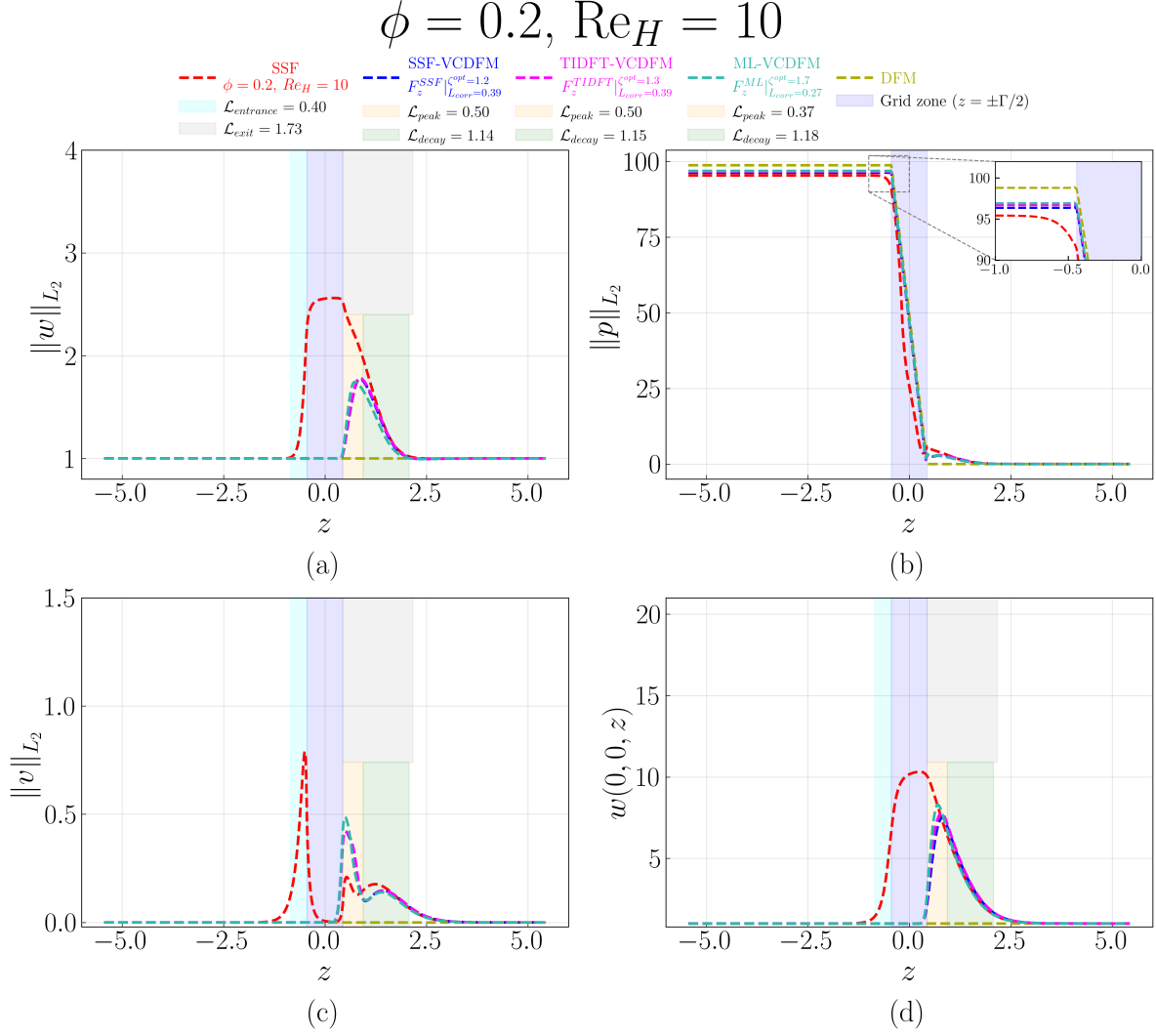


Figure D.135: Streamwise evolution of (a)  $\|w\|_{L_2}$ , (b)  $\|p\|_{L_2}$ , (c)  $\|v\|_{L_2}$ , and (d) centerline velocity  $w(0,0,z)$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 10$ . The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based  $L_{\text{peak}}$  and  $L_{\text{decay}}$  are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

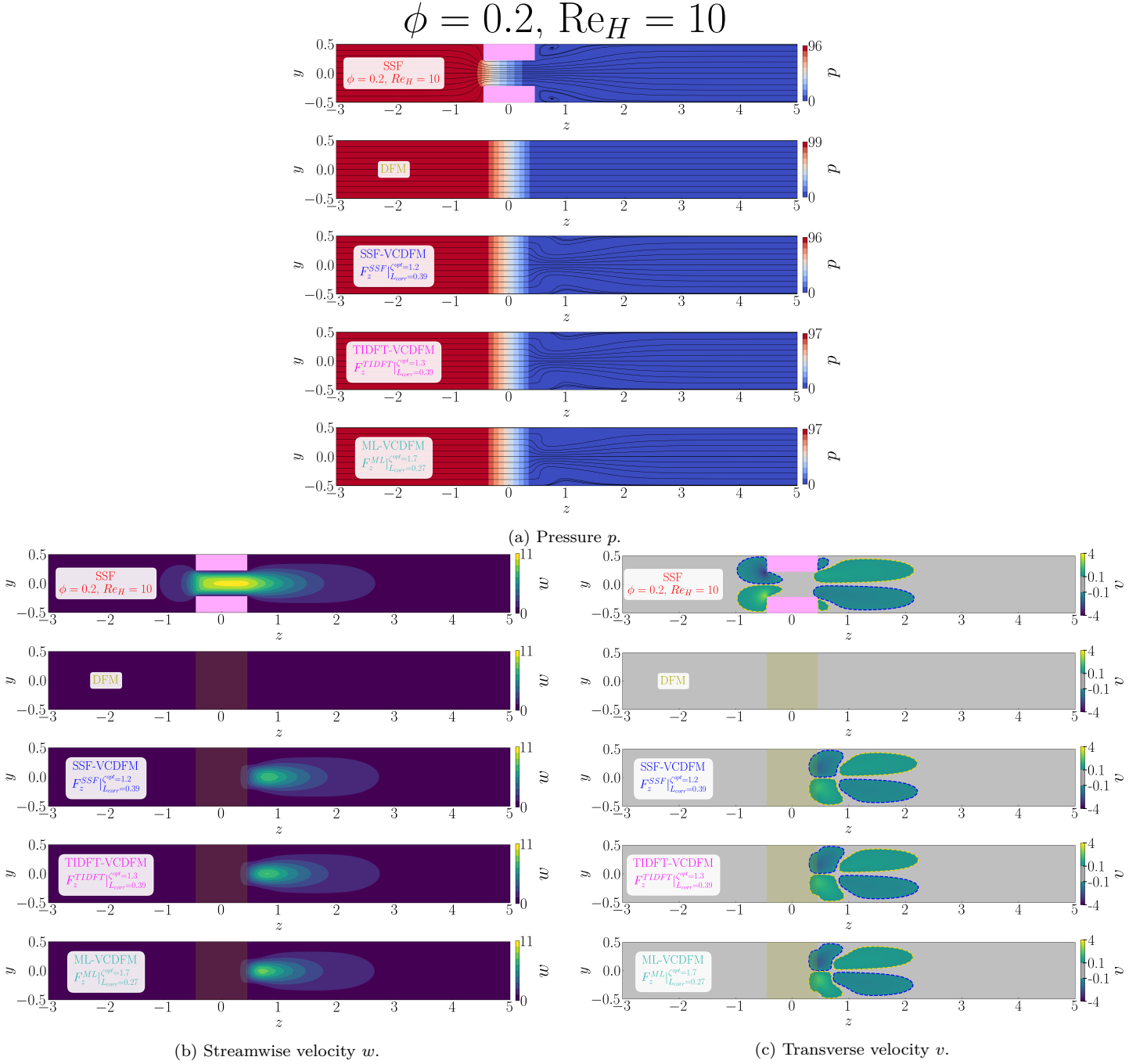


Figure D.136: Flow organization in the streamwise  $(z, y)$ -plane for the  $1 \times 1$  SRG at  $\phi = 0.2$ ,  $\text{Re}_H = 10$ , and  $\alpha = 0^\circ$ . The blocks show (a) pressure  $p$  with streamlines, (b) streamwise velocity  $w$ , and (c) transverse velocity  $v$ . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

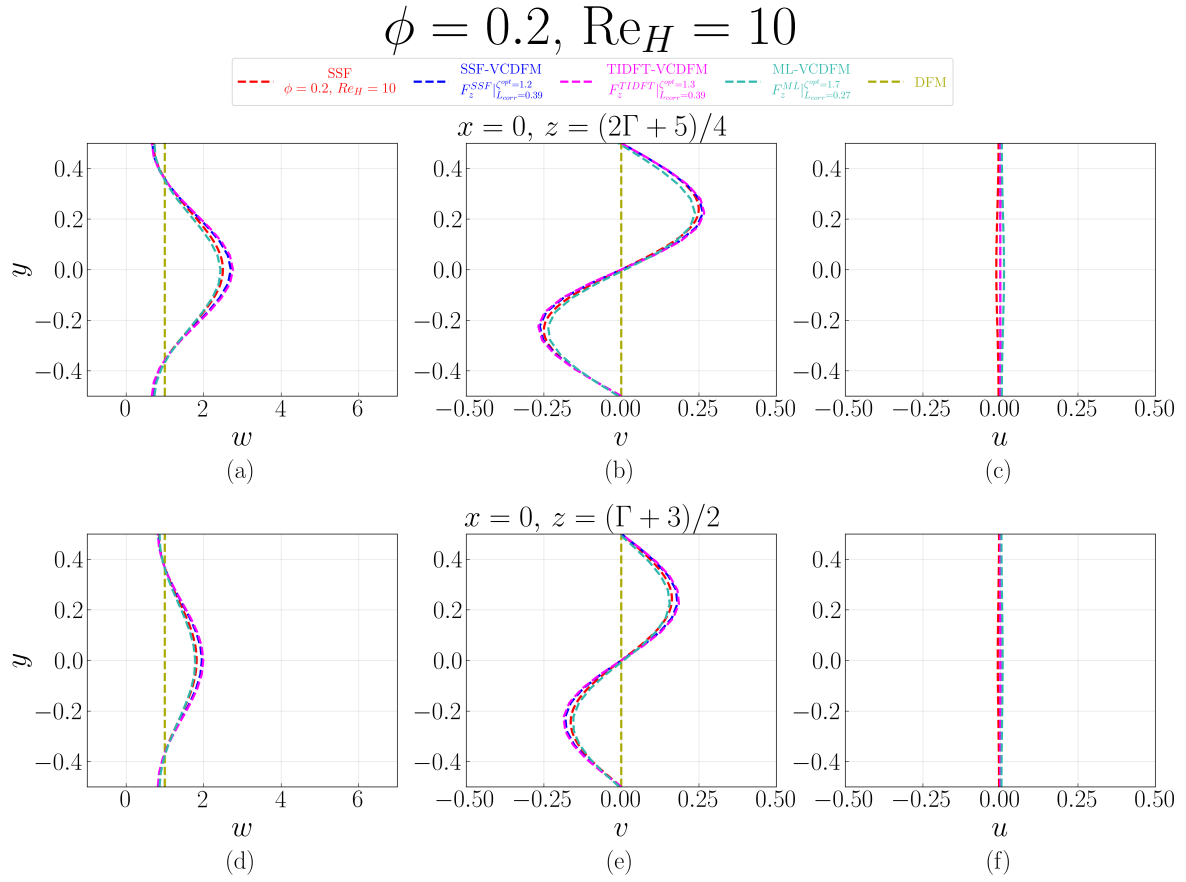


Figure D.137: Velocity profiles at  $x = 0$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 10$ , comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of  $w(y)$ ,  $v(y)$ , and  $u(y)$  are shown at  $z = (2\Gamma + 5)/4$  in (a–c) and  $z = (\Gamma + 3)/2$  in (d–f).

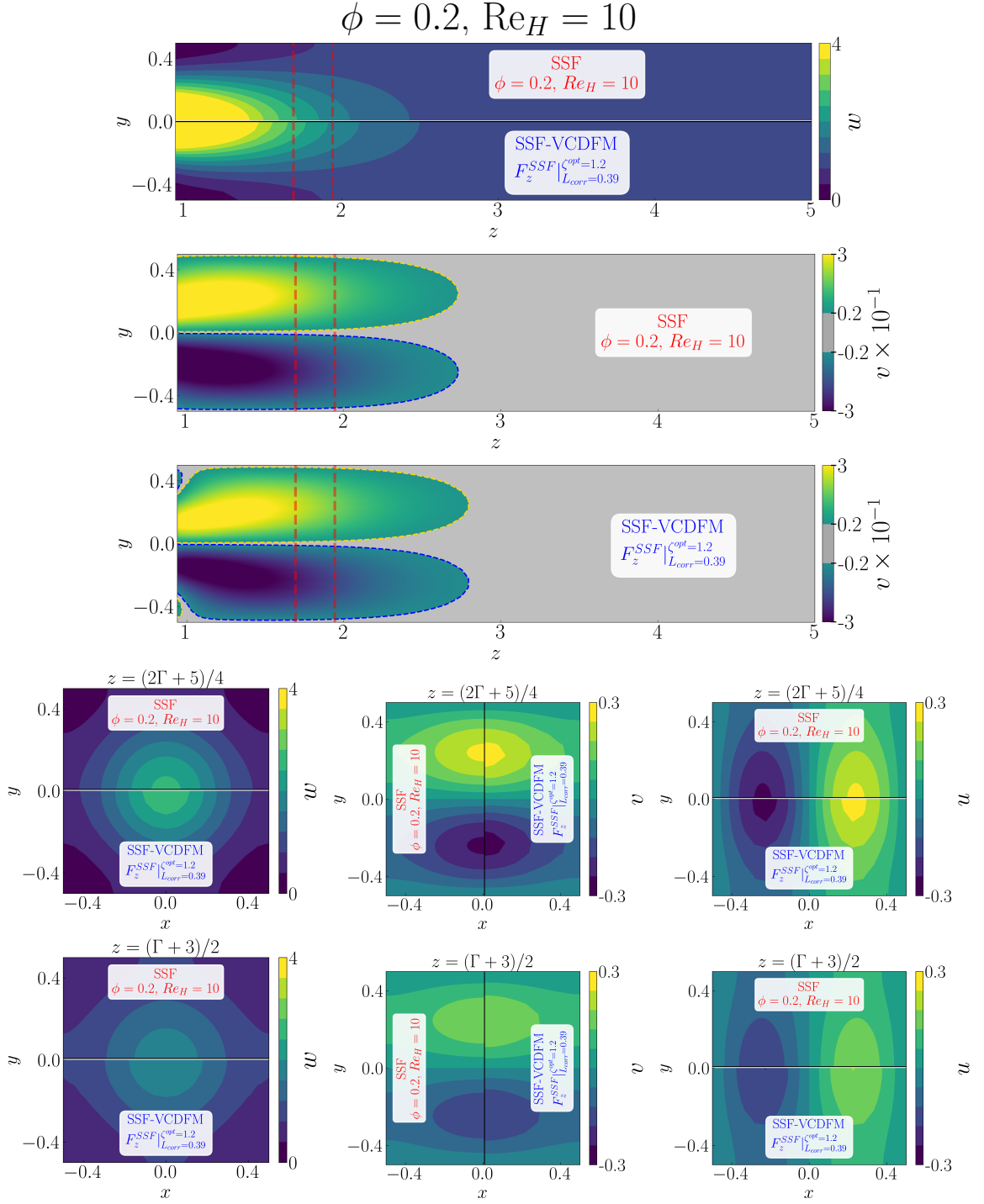


Figure D.138: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using  $F_z^{SSF}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $Re_H = 10$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (2\Gamma + 5)/4$  in the first row and  $z = (\Gamma + 3)/2$  in the second row.

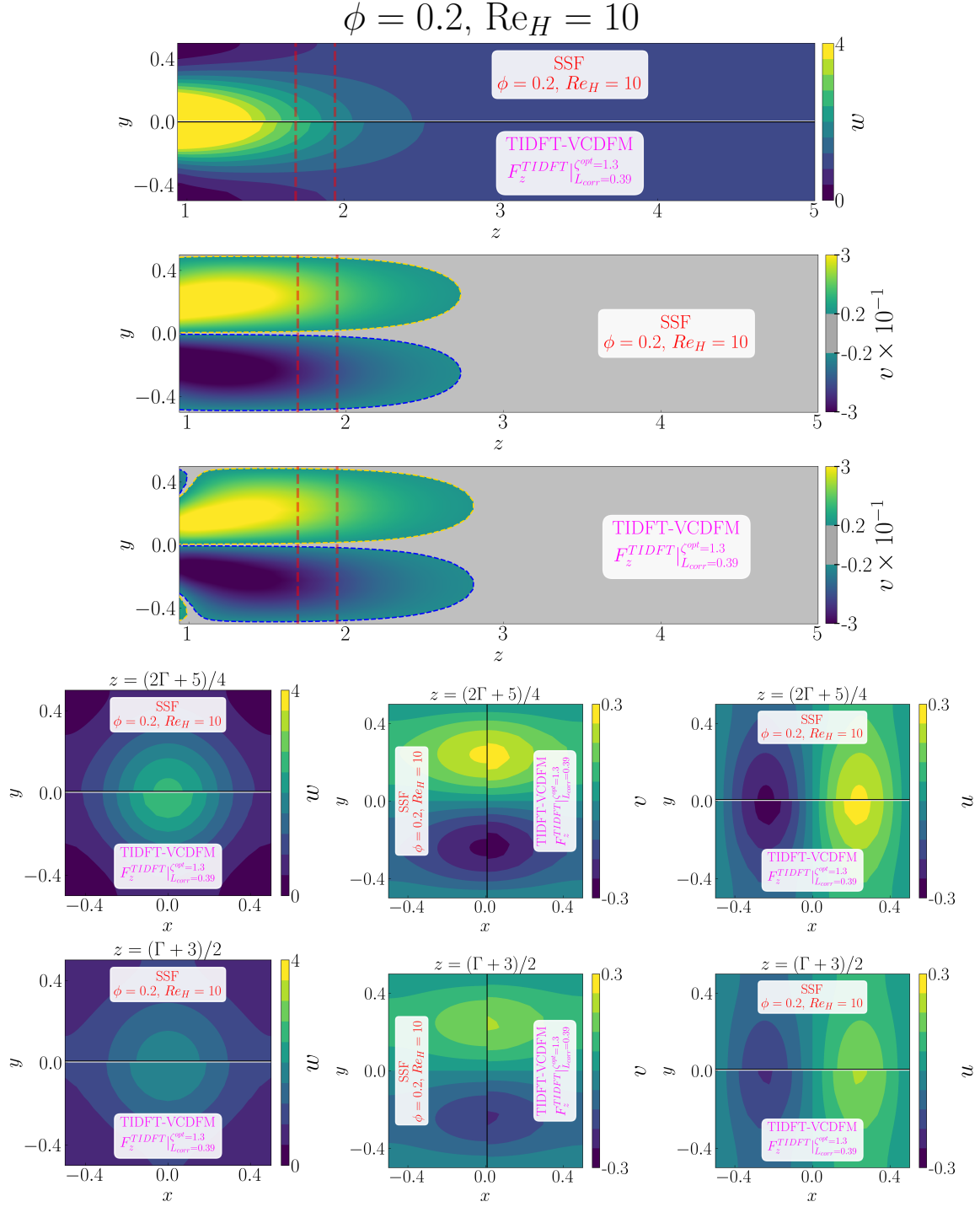


Figure D.139: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using  $F_z^{TIDFT}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $Re_H = 10$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (2\Gamma + 5)/4$  in the first row and  $z = (\Gamma + 3)/2$  in the second row.

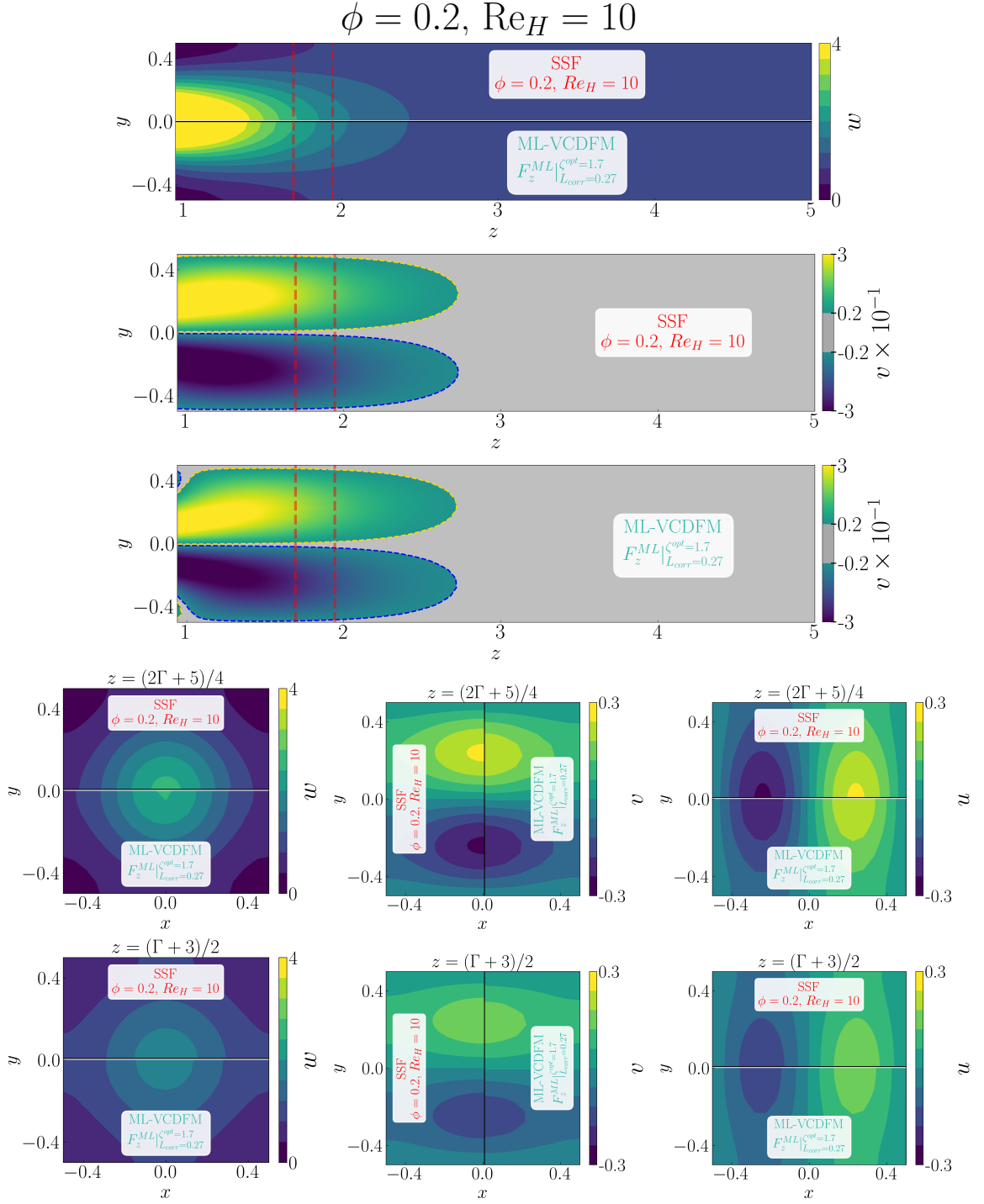


Figure D.140: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using  $F_z^{ML}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $Re_H = 10$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (2\Gamma + 5)/4$  in the first row and  $z = (\Gamma + 3)/2$  in the second row.

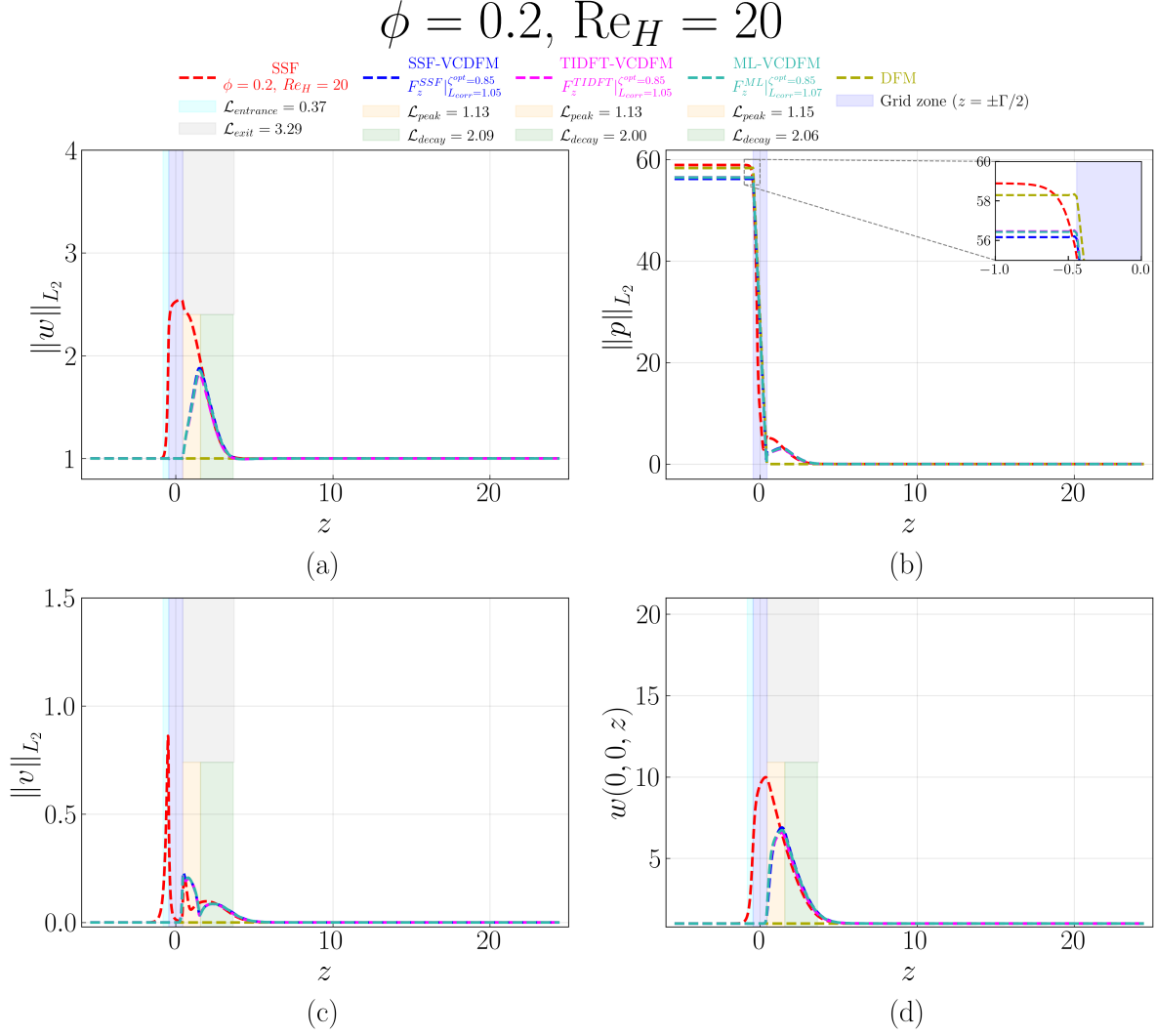


Figure D.141: Streamwise evolution of (a)  $\|w\|_{L_2}$ , (b)  $\|p\|_{L_2}$ , (c)  $\|v\|_{L_2}$ , and (d) centerline velocity  $w(0,0,z)$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 20$ . The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based  $\mathcal{L}_{\text{peak}}$  and  $\mathcal{L}_{\text{decay}}$  are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

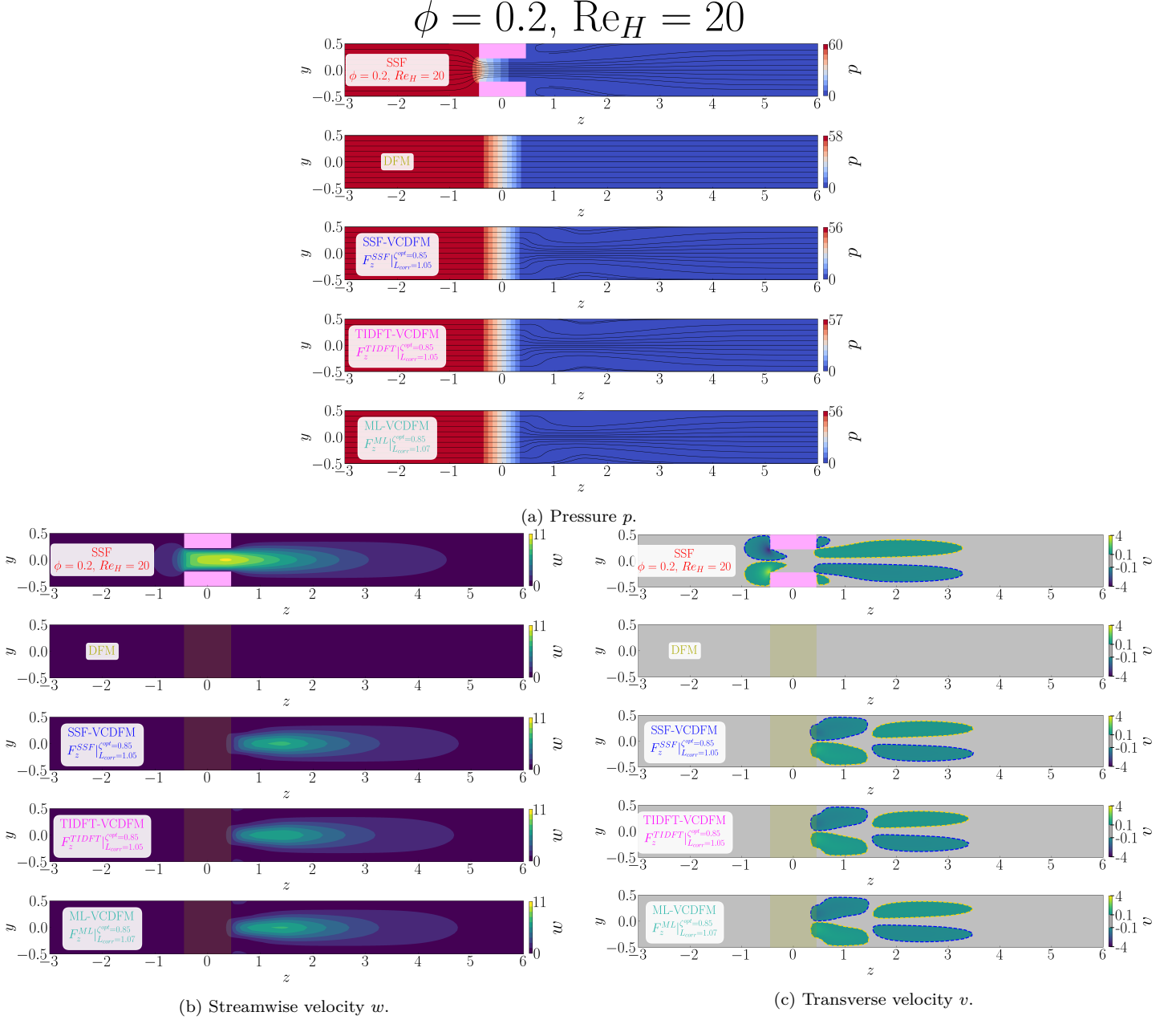


Figure D.142: Flow organization in the streamwise  $(z, y)$ -plane for the  $1 \times 1$  SRG at  $\phi = 0.2$ ,  $Re_H = 20$ , and  $\alpha = 0^\circ$ . The blocks show (a) pressure  $p$  with streamlines, (b) streamwise velocity  $w$ , and (c) transverse velocity  $v$ . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

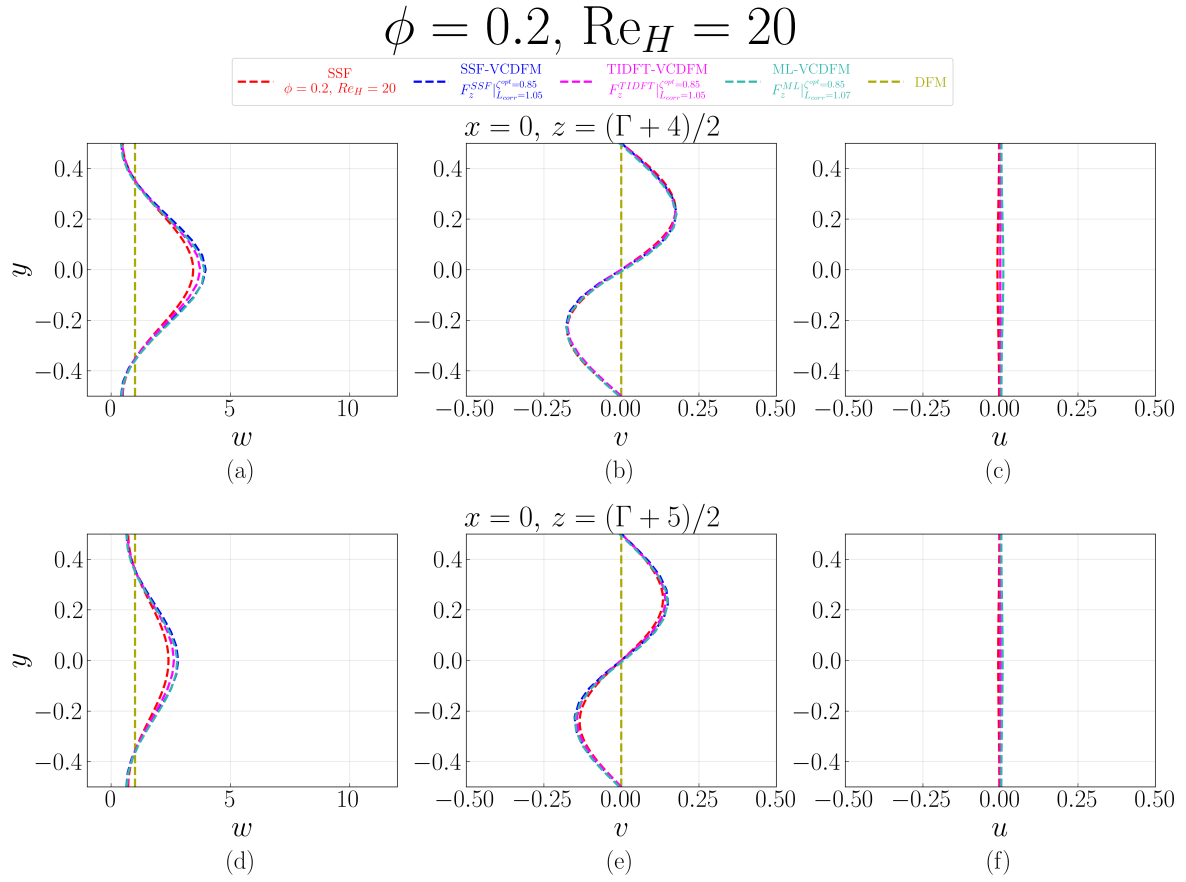


Figure D.143: Velocity profiles at  $x = 0$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 20$ , comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of  $w(y)$ ,  $v(y)$ , and  $u(y)$  are shown at  $z = (\Gamma + 4)/2$  in (a–c) and  $z = (\Gamma + 5)/2$  in (d–f).

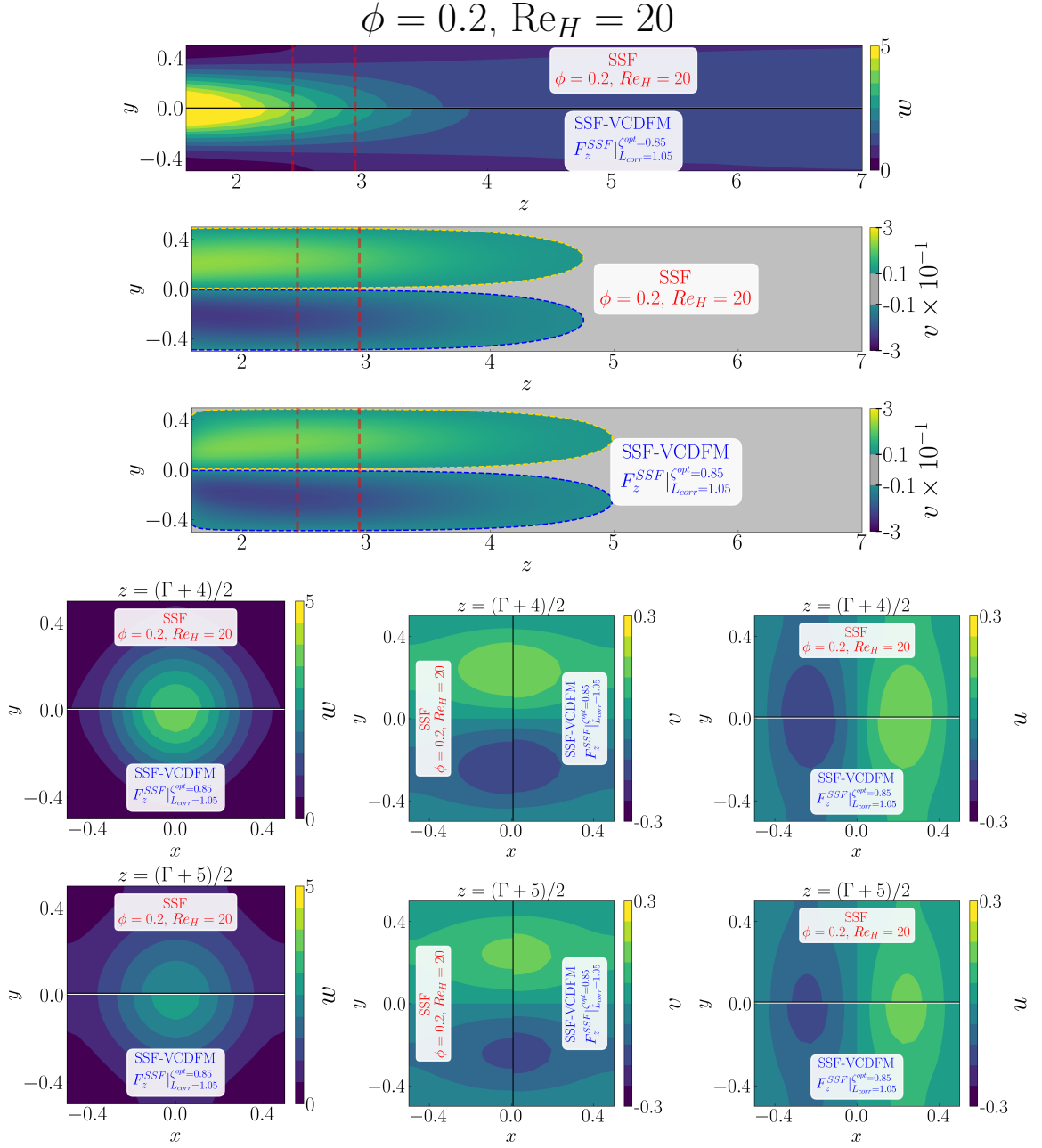


Figure D.144: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using  $F_z^{\text{SSF}}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $\text{Re}_H = 20$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (\Gamma + 4)/2$  in the first row and  $z = (\Gamma + 5)/2$  in the second row.

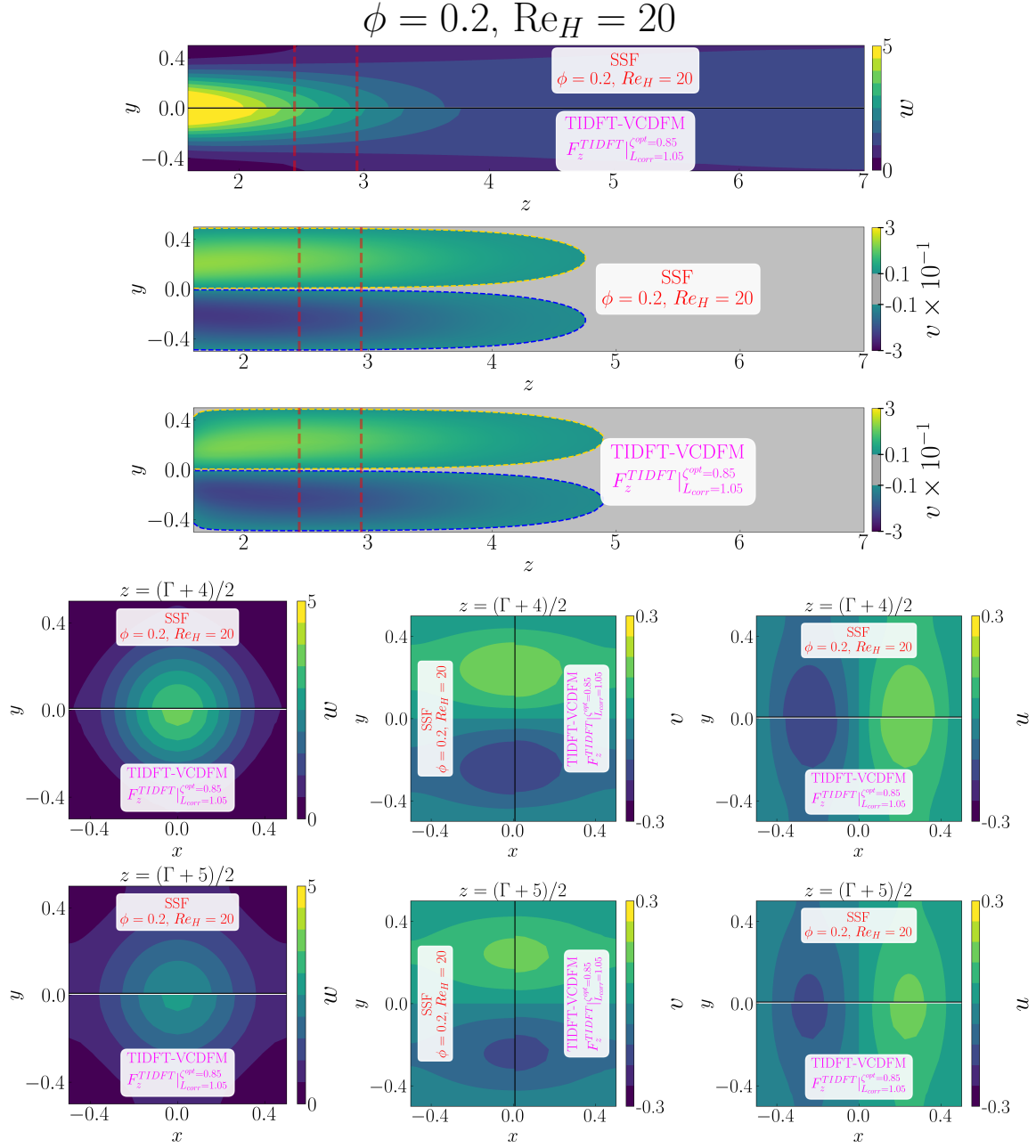


Figure D.145: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using  $F_z^{TIDFT}$  for the  $1 \times 1$  SRG at  $\phi = 0.2$  and  $Re_H = 20$ . The upper panels compare the streamwise velocity  $w$  and transverse velocity  $v$  in the central  $(z, y)$ -plane, restricted to the correction region. The lower panels compare the corresponding transverse  $(x, y)$ -plane distributions of  $w$ ,  $v$ , and  $u$ , from left to right, at  $z = (\Gamma + 4)/2$  in the first row and  $z = (\Gamma + 5)/2$  in the second row.